

Week 2

Describing Data

What an average reveals, and what it hides

FUNDAMENTAL DATA SCIENCE · LECTURE



Three checks before an average

Before you calculate, know what each row and each value represents.

What is one row?

One student, one registration, or one student-week?

The unit determines the denominator.

What is the unit?

Points, minutes, a count, or a category?

Arithmetic needs a meaningful measurement scale.

How much data?

Report valid n and what is missing.

A mean without a denominator can mislead.

START HERE

One row is one _____. Values are measured in _____. We have _____ valid rows.

Last week: what a number is allowed to mean

A concern is not a question

until it names a population, a variable, a period and a decision

A link is not a cause

Students who log in more may earn higher grades. That does not prove logging in caused the grades.

A row is a choice

4,665 registrations are not 4,665 students

Count the right thing

One student may have several registrations. Count students once for a student-level rate.

This week: an average can hide missing answers, large differences, and separate groups.

One row means something different in each table

Illustrative rows: notice what the same student ID counts in each file.

students.csv



student_id	programme	entry_score
100003	Computer Science	68

enrolments.csv



student_id	course_code	final_grade
100003	DS100	72

activity.csv



student_id	week	minutes
100003	1	44

REMEMBER

One student can have several course rows and many weekly activity rows.

Three questions before we describe data

Ask who was counted, what is typical, and how much the values differ.

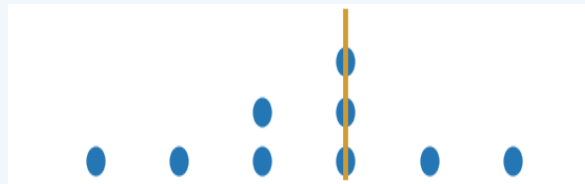
Who was counted?



Some students may skip a survey question.

An average then uses fewer answers.

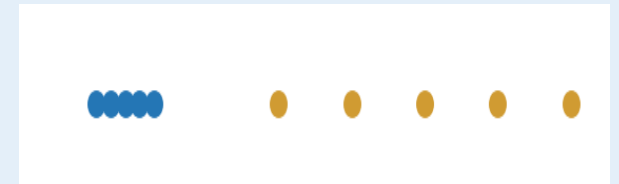
What is typical?



Where do most scores sit?

We will compare the mean and median.

How different?



Are scores close together or far apart?

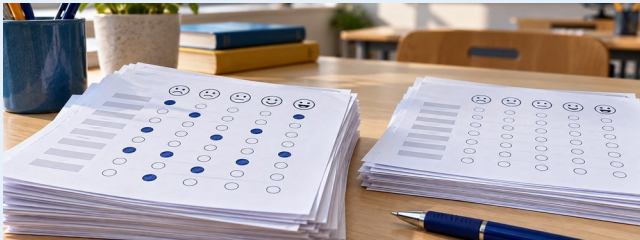
A picture can show the difference.

Who is included in this average?

The class has 1,200 students, but only 916 gave a satisfaction score.

Answered

$916 \div 1,200 = 76.3\%$ answered.



No answer recorded

$284 \div 1,200 = 23.7\%$ have no recorded answer.

Report clearly

Mean of recorded answers = sum of 916 scores $\div$ 916 = 3.5. We cannot assume the 284 missing students felt the same.

QUESTION

Which students are missing, and could that difference change the conclusion?

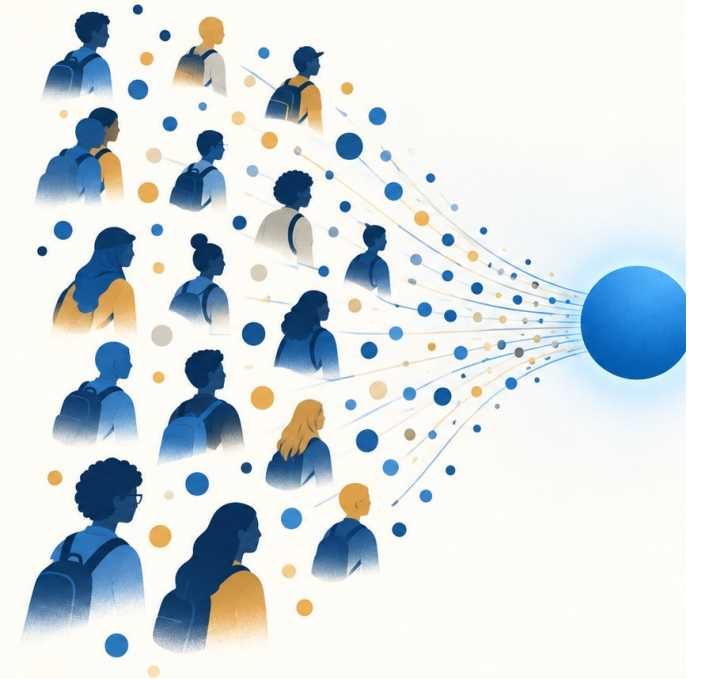
Is one average enough to plan tutoring?

The Dean wants to decide whether incoming students need extra support.

Think for 30 seconds

**The average entry score is 65.9.
What else should the Dean check
before planning tutoring?**

For example: how many students scored low, and do programmes differ?



PURPOSE

A mean alone cannot show how many students might need help.

PART ONE • AVERAGES

Many scores. One number.

An average helps us compare groups. It can also hide who needs support.



What compression erases

One class average can hide people, patterns, and how the data was collected.

Individuals

A mean of 66 does not show which students scored 35 or 95.



Structure

Two classes might have different score patterns, even with the same mean.



Context

A score means more when we know who was tested and when.



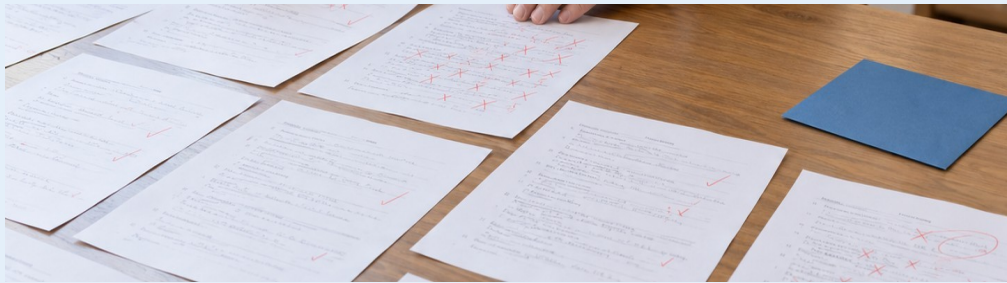
DECISION TEST

Before using one number, ask which hidden details could change your decision.

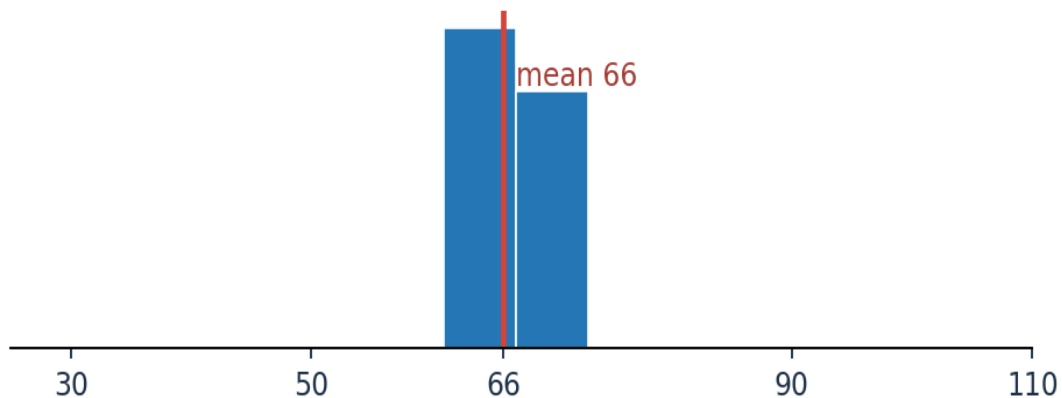
One class average hides individual scores

What you gain

A single number makes it easy to compare classes.

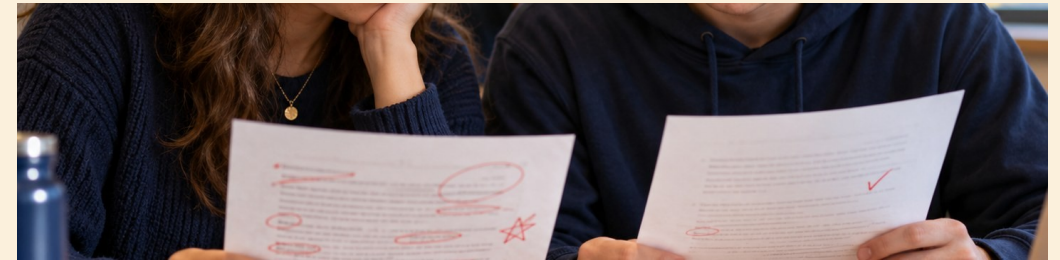


Class A: scores close together

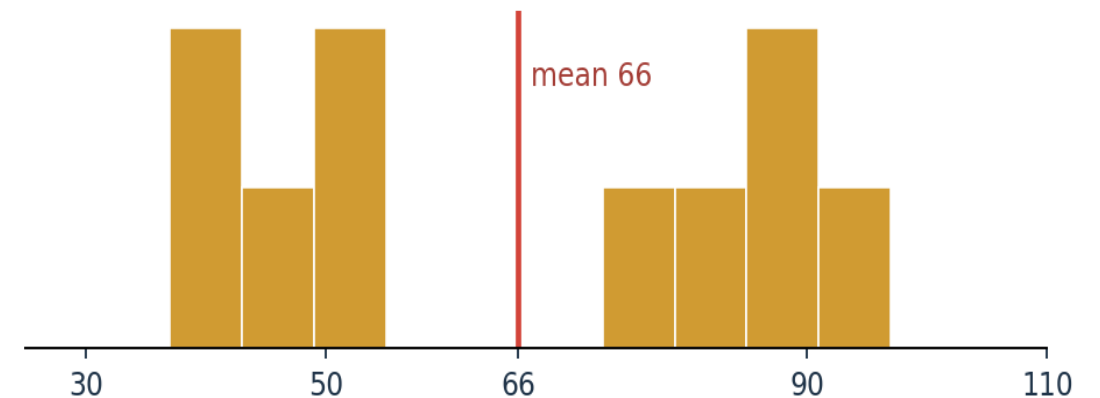


What you lose

The same mean can come from very different score patterns.



Class B: scores split into two groups



Three answers to “what is typical?”

Mean

add them all, divide by how many

uses every value — and is moved by every value

Median

sort them, take the middle one

ignores how far away the extremes are

Mode

the most common value

the only one that works for categories

In our data

entry_score
mean 65.86
median 65.90

Almost identical — which itself tells you something about the shape.

When they diverge, that divergence is information, not a nuisance.

CHECK

A programme name (such as Computer Science) is a category. Which programme occurs most often?

The mean is a balance point

The deviations from the mean add to zero. That is why every value can move it.

52, 58, 63, 71, 76 → mean = 64

Check the balance

Below 64: $-12 - 6 - 1 = -19$

Above 64: $+7 + 12 = +19$

Why the mean moves

Change 76 to 760 and the mean becomes 200.8. One value changed, so the balance point moved with it.

WHEN TO USE / AVOID

Use the mean for a fairly even score pattern or for totals. Avoid it as a “typical” score when an error such as 760 pulls it far upward.

The median is a position

Sort the values, then find the middle. Magnitude beyond the middle does not move it.

52, 58, 63, 71, 76 → median = 63

Extreme change

Replace 76 by 760.
Median remains 63.
Mean becomes 200.8.

Use the contrast

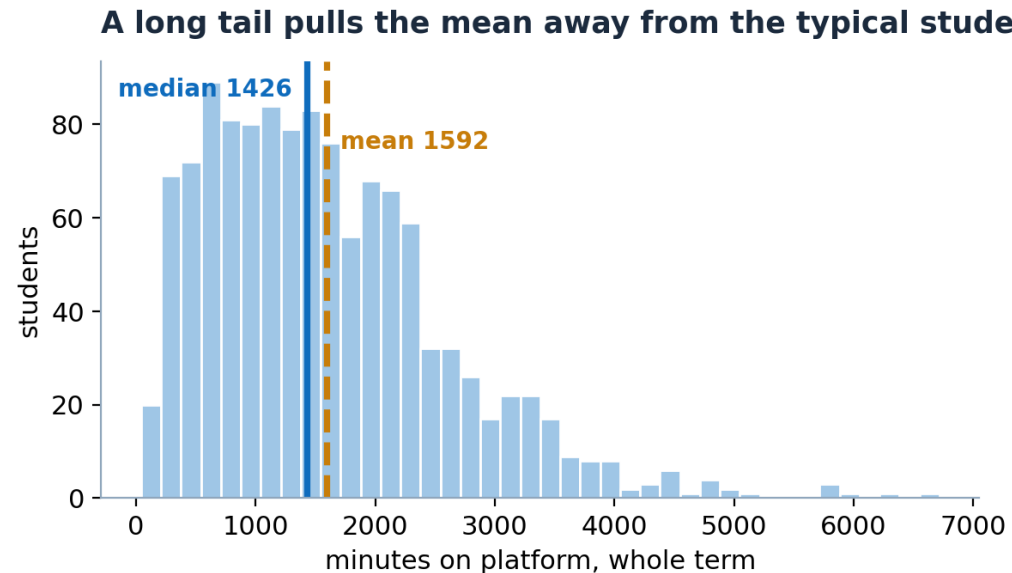
Mean and median disagree when a tail or extreme values pull the balance point away from the middle observation.

WHEN TO USE / AVOID

Use the median for a typical student when a few values are extreme. Avoid it alone for total workload or when two groups have different patterns.

A long tail pulls the mean away from the typical student

Total minutes on the platform, per student, whole term. Same 1,200 students as last week.



Median 1,426 minutes describes a typical student. Mean 1,592 minutes is higher because a few students use the platform much more.

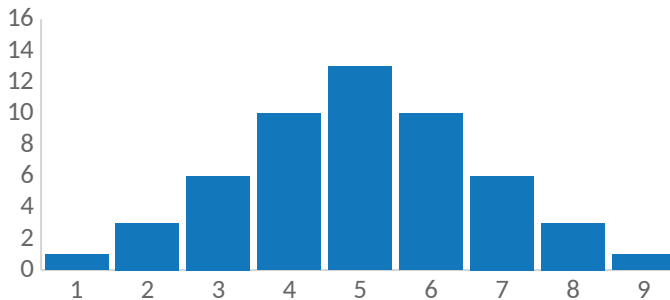
★ WHICH WOULD YOU REPORT?

For a typical student, use the median. For total platform capacity, use the mean or total minutes. Why?

Distribution shape guides the summary

For a typical value, choose a centre and a measure of spread together.

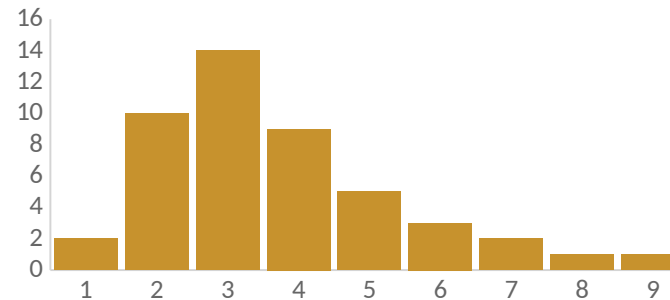
Approximately symmetric



Mean + SD

One peak and no strong outliers.
Mean describes centre, SD spread.

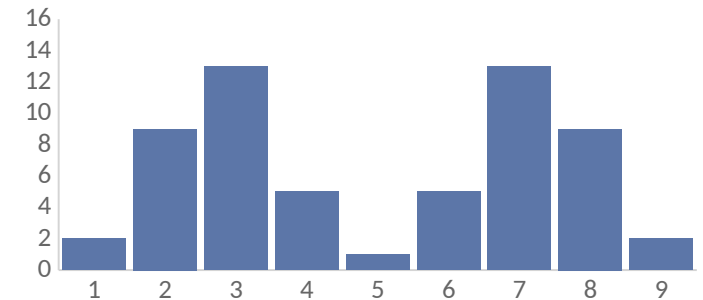
Skewed or with outliers



Median + IQR

A long tail pulls the mean.
Median and IQR are less sensitive.

Two or more peaks



Group summaries

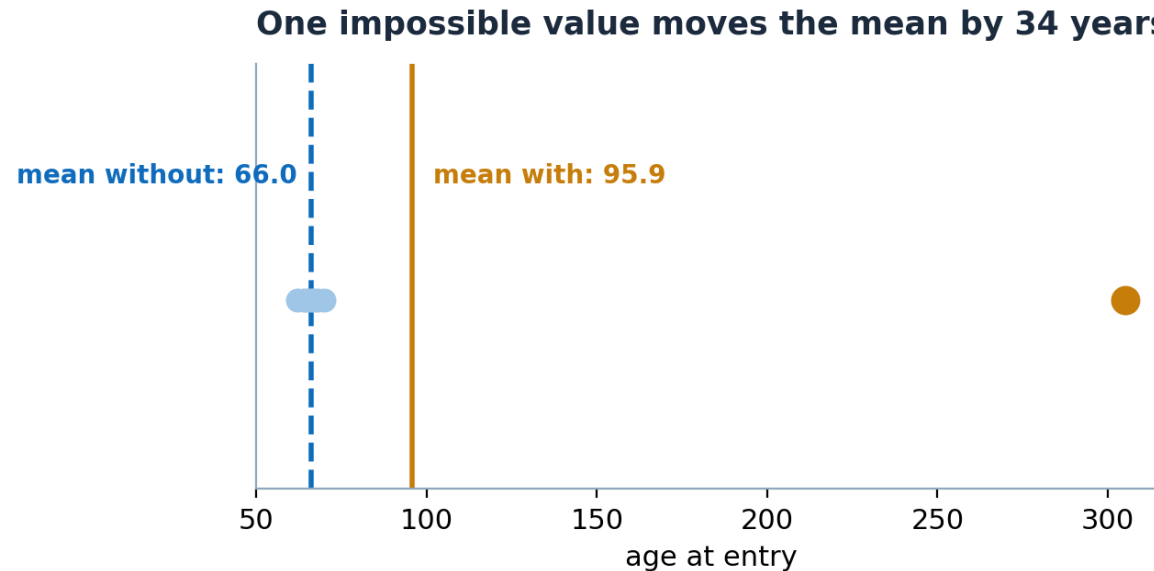
One centre can hide distinct groups.
Show the plot and describe each group.

Illustrative shapes. The skewed example has a right tail. The same advice applies to a long left tail.

Coming next: SD measures spread. IQR measures the width of the middle half.

One impossible value moves the mean by 34 years

Seven real ages at entry, plus one value of 305 — which is really in `students_dirty.csv`, and which you will meet in Week 3.



The median of the eight values is 66.5, barely moved. The mean is wrecked.

★ THE LESSON IS NOT “USE THE MEDIAN”

It is: compute both, and if they disagree, go and look at the data before choosing.

Quartiles

The middle 50%

Three cut points in sorted data show where a value sits.

Q1 Q2 / MEDIAN Q3



Quartiles: three cut points in sorted data

Quartiles divide an ordered dataset into four groups of approximately equal size.

Q1

25% position

Q2 / median

50% position

Q3

75% position

Example: 8 sorted scores

lower four

upper four

35	52	58	62	66	70	74	82
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1. Q2: middle of all eight

$$(62 + 66) \div 2 = 64$$

2. Q1: middle of lower four

$$(52 + 58) \div 2 = 55$$

3. Q3: middle of upper four

$$(70 + 74) \div 2 = 72$$

Method: median of halves. Software may interpolate differently, so always state the quartile method.

The five-number summary

Five numbers describe the lowest value, the quarter positions, and the highest value.

Number	What it tells us	8-score example	Full class (1,200)
Minimum	Smallest observed value	35	35.0
Q1	Lower quarter: about 25% at or below it	55	58.3
Median (Q2)	Middle: about 50% at or below it	64	65.9
Q3	Upper quarter: about 75% at or below it	72	73.6
Maximum	Largest observed value	82	99.0

Example interpretation

Class Q1 = 58.3 means about one quarter of students scored 58.3 or below.

IQR: subtract Q1 from Q3

The interquartile range (IQR) is the width of the middle half of the scores.

Practice list: $72 - 55 = 17$. Now calculate it for all 1,200 scores:

35.0

min

58.3

Q1

65.9

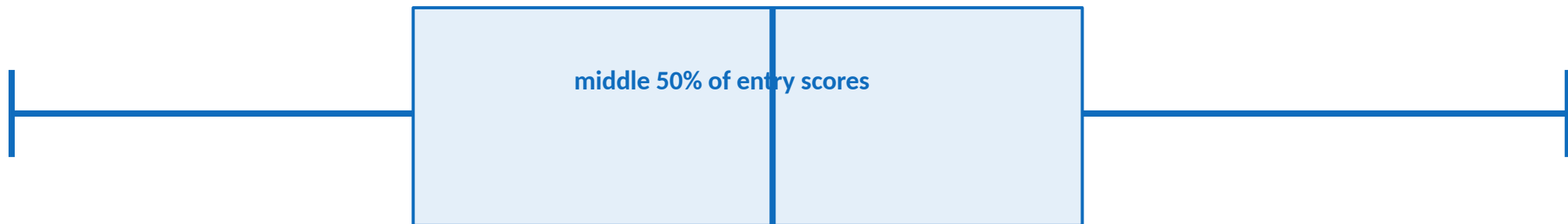
median

73.6

Q3

99.0

max



Course IQR = $Q3 - Q1 = 73.6 - 58.3 = 15.3$ points

INTERPRET

Half the entry scores lie from 58.3 to 73.6. The box is 15.3 score points wide.

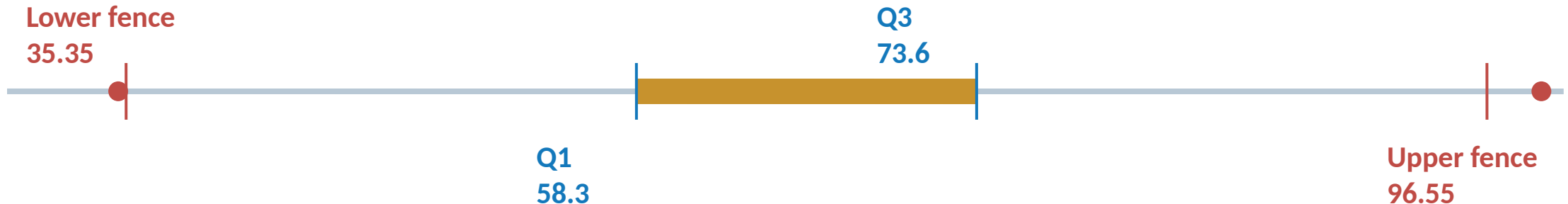
Outlier fences: 1.5 IQR beyond the box

$Q1 = 58.3$ and $Q3 = 73.6$, so $IQR = 73.6 - 58.3 = 15.3$ points.

$$1.5 \times IQR = 1.5 \times 15.3 = 22.95$$

$$\begin{aligned}\text{Lower fence} &= Q1 - 1.5 \times IQR \\ &= 58.3 - 22.95 = 35.35\end{aligned}$$

$$\begin{aligned}\text{Upper fence} &= Q3 + 1.5 \times IQR \\ &= 73.6 + 22.95 = 96.55\end{aligned}$$



$35 < 35.35$ and $99 > 96.55$: flag both values for investigation.

A fence is a threshold. A flagged value may be valid and should not be deleted automatically.

Flag, investigate, document

Unusual values have several possible explanations. A deletion rule cannot distinguish them.

Wrong value

Age 305 in students_dirty.csv

Check the original record and correct or remove only with evidence.

Valid extreme

6,717 total platform minutes

Could be unusually high but valid engagement. Inspect, do not erase.

Different population

A value may be ordinary inside a subgroup

Consider course, programme, period, and measurement process.

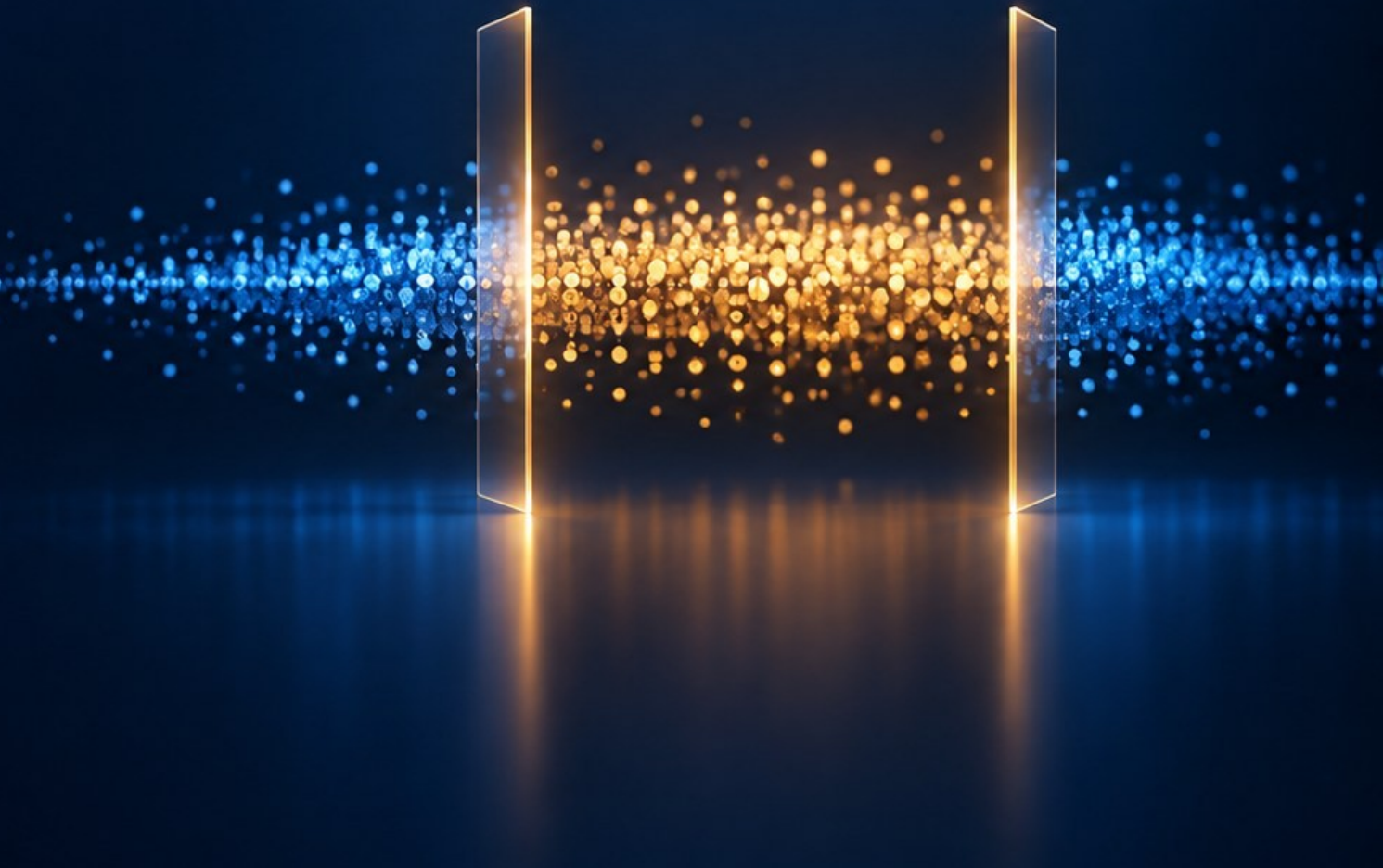
AUDIT TRAIL

Every correction or exclusion needs a reproducible rule and a written reason.

Spread

How different are the scores?

A mean locates the centre.
Spread describes the distances
between the values.



Standard deviation: a worked example

Sample scores: 2, 4, 6, 8 points. Mean = $(2 + 4 + 6 + 8) \div 4 = 5$ points.

Score x	$x - \text{mean}$	$(x - \text{mean})^2$
2	-3	9
4	-1	1
6	+1	1
8	+3	9
Total	0	20

3. Divide and take the square root

Sample variance $s^2 = 20 \div (4 - 1) = 6.67$ points²

1. Subtract the mean

This measures each score's distance and direction from the centre.

2. Square, then add

Squaring stops positive and negative distances from cancelling.

$$\text{SD} = \sqrt{6.67} = 2.58 \text{ points}$$

Larger SD means more spread.

We explain the $n - 1$ divisor shortly.

Three score sets with the same mean of 66

The average stays fixed, but the students' experience changes as the values spread out.

Class A

64 65 66 67 68

mean = 66 SD = 1.58

Class B

54 60 66 72 78

mean = 66 SD = 9.49

Class C

34 47 66 85 98

mean = 66 SD = 26.32

Same mean, very different spread. A centre without a spread does not describe the group.

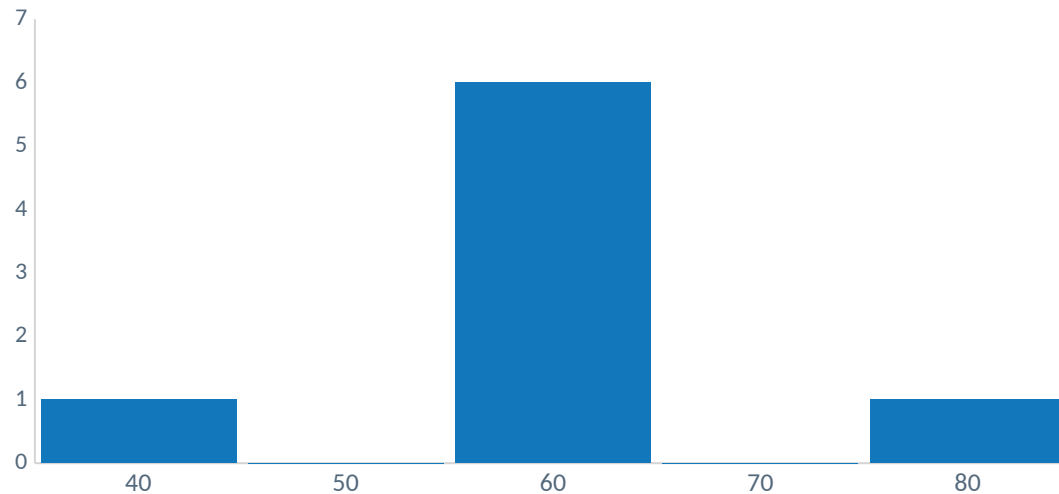
SAY IT

A mean alone hides how different the scores are.

The same mean and SD can hide different shapes

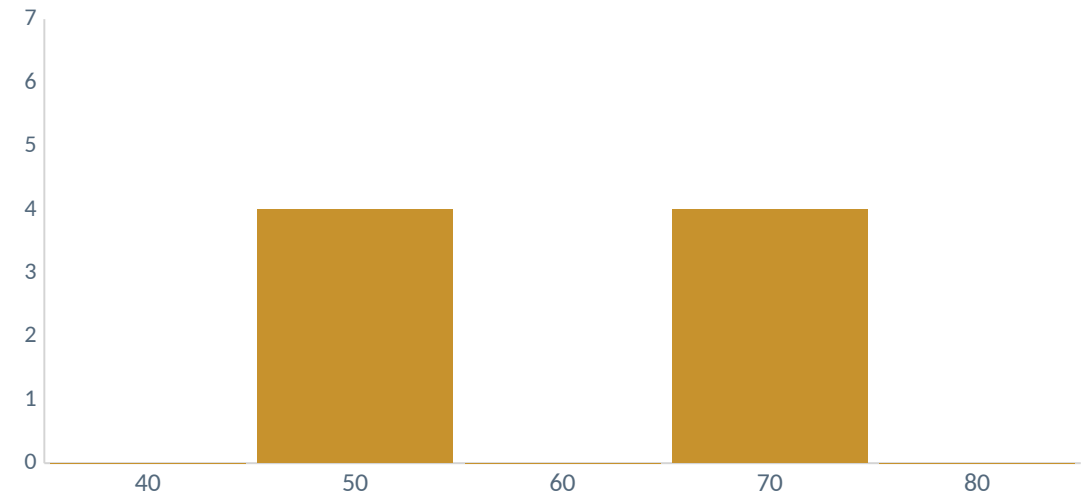
Both groups have $n = 8$, mean = 60 and sample standard deviation = 10.69 points.

Group A: most scores at the centre



40, 60, 60, 60, 60, 60, 60, 80

Group B: two separated groups



50, 50, 50, 50, 70, 70, 70, 70

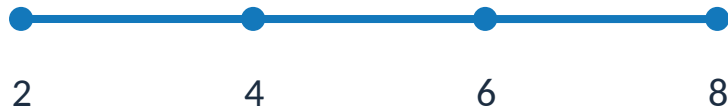
For each group: $\sum(x - 60)^2 = 800$, so $s = \sqrt{(800 \div 7)} = 10.69$.

A plot adds information that a mean and SD cannot contain.

Four measures of spread, one example

Scores: 2, 4, 6, 8. Mean = 5. Q1 = 3 and Q3 = 7 using the median of halves.

Range



$$\text{max} - \text{min} = 8 - 2 = 6 \text{ points}$$

Sample variance

Distances: -3, -1, +1, +3

Squares: 9, 1, 1, 9

$$s^2 = (9 + 1 + 1 + 9) \div (4 - 1) \\ = 6.67 \text{ points}^2$$

Interquartile range (IQR)



$$Q3 - Q1 = 7 - 3 = 4 \text{ points}$$

Standard deviation (SD)

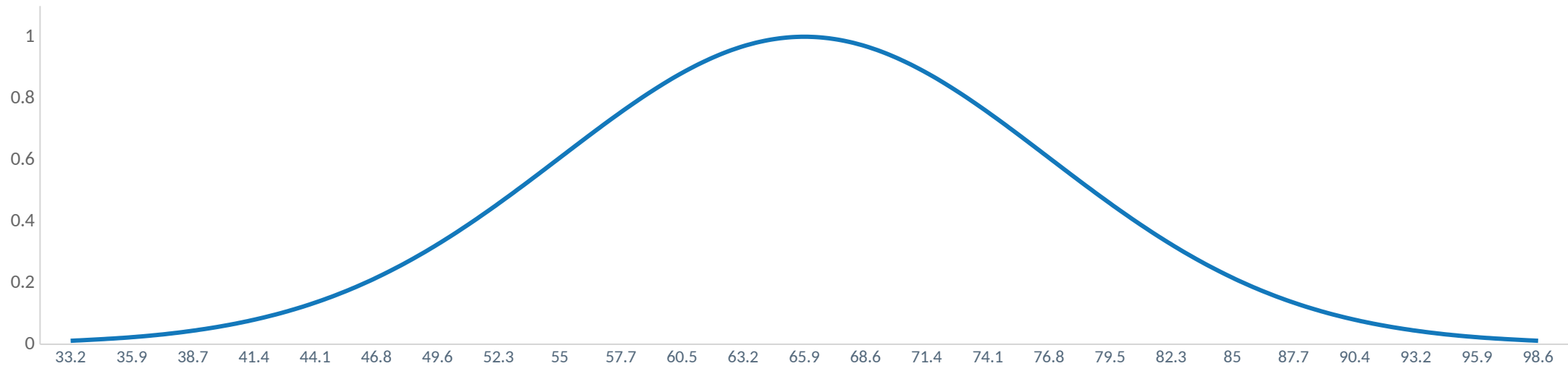
Variance 6.67 points²

Take the square root

$$s = \sqrt{s^2} = \sqrt{6.67} \\ = 2.58 \text{ points}$$

Distances from the mean on a bell-shaped curve

Normal-model illustration: mean = 65.9 points and SD = 10.9 points.



Interval	Calculation	Approximate share
Within 1 SD	$65.9 \pm 10.9 = 55.0 \text{ to } 76.8$	68%
Within 2 SD	$65.9 \pm 21.8 = 44.1 \text{ to } 87.7$	95%

These percentages need an approximately normal shape. Symmetry alone is not enough.

Why variance has squared units

Use the same scores: 2, 4, 6, 8 points, with a mean of 5 points.

Measure	Numerical calculation	Unit
Range	$8 - 2 = 6$	points
Interquartile range	$7 - 3 = 4$	points
Sample variance	$(9 + 1 + 1 + 9) \div 3 = 6.67$	points ²
Standard deviation	$\sqrt{6.67} = 2.58$	points

Squaring changes the unit

$$(2 \text{ points} - 5 \text{ points})^2 = (-3 \text{ points})^2 \\ = 9 \text{ points}^2$$

The square root restores it

$$\sqrt{(6.67 \text{ points}^2)} = 2.58 \text{ points}$$

Range, IQR and SD all use the original unit.

How one extreme value changes each measure

Two lists of 9 scores. Only the largest score changes, from 8 to 80.

A: 2, 3, 4, 4, 5, 6, 6, 7, 8

B: 2, 3, 4, 4, 5, 6, 6, 7, 80

Measure	List A	List B	Why it changes
Range	$8 - 2 = 6$	$80 - 2 = 78$	Uses the smallest and largest values
Interquartile range (IQR)	$6.5 - 3.5 = 3$	$6.5 - 3.5 = 3$	The middle half stays the same
Standard deviation (SD)	1.94	25.17	Uses every distance from the mean

Use range for observed limits, IQR for a robust middle spread, and SD for spread around the mean.

Why sample variance divides by $n - 1$

A sample contains some observations from a larger population that we want to describe.

Four scores: 2, 4, 6, 8

Sample mean = 5

Deviations: -3, -1, +1, +3

Only three deviations are free

The deviations must sum to zero.
Once -3, -1 and +1 are known,
the last deviation must be +3.

Question	Divisor	Variance
Describe these four values as the full population	$n = 4$	$20 \div 4 = 5$
Estimate a larger population's variance from this sample	$n - 1 = 3$	$20 \div 3 = 6.67$

Using the sample mean makes variance too small on average. Dividing by $n - 1$ corrects that bias.

Software defaults must be named

Two valid formulas can produce different values when they answer different questions.

```
s = students['entry_score']  
  
s.std()           # pandas: n - 1  
np.std(s)         # NumPy: n  
np.std(s, ddof=1) # matches pandas
```

Why the difference

pandas defaults to the sample standard deviation. NumPy defaults to a population-style divisor.

What to write

State the library and ddof setting whenever another analyst could reproduce a different value.

LAB 2

Students compute both versions by hand and in code. The mismatch is the lesson, not a bug.

One column

Two different centres

Platform minutes:

Mean = 1,592

Median = 1,426

What does the distribution look like?



Shape

Patterns an average cannot show

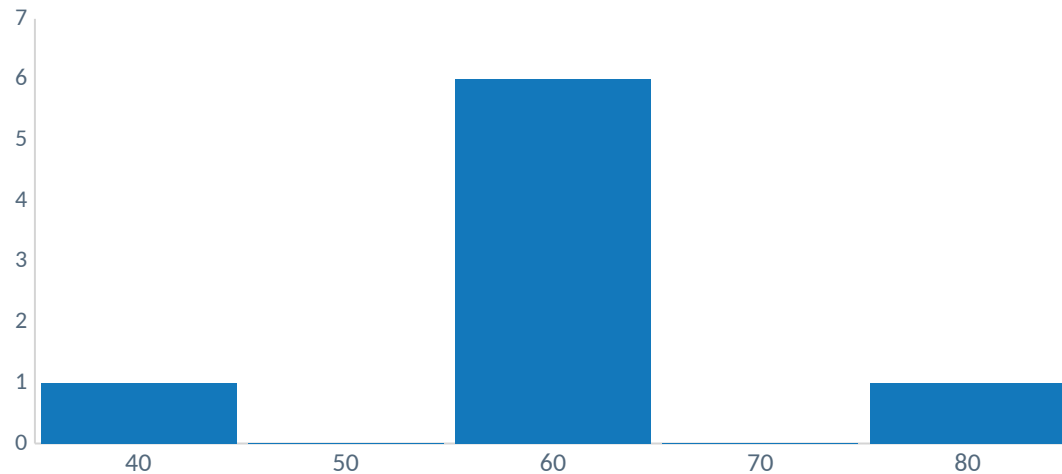
A distribution can have a long tail or separate groups. A histogram makes these patterns visible.



Which class needs two levels of tutoring?

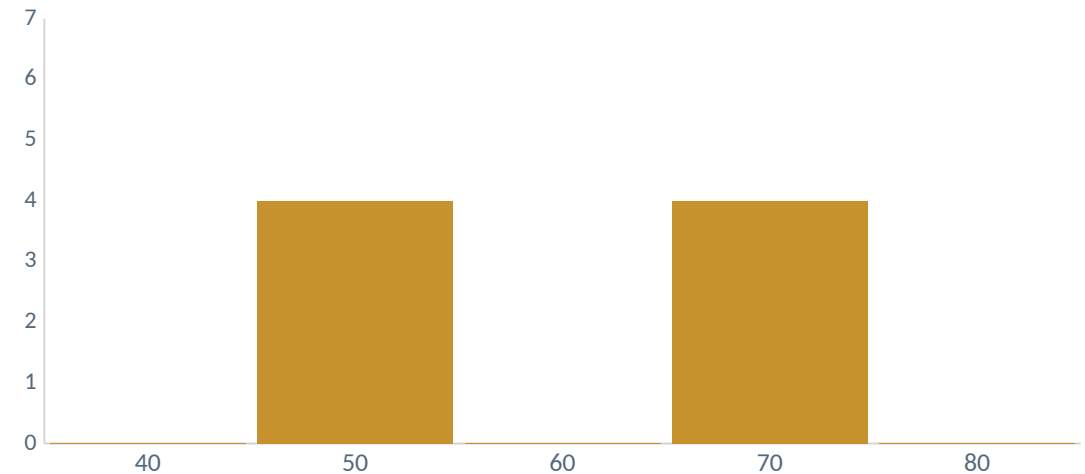
Each class has 8 students, a mean score of 60 and a sample SD of 10.69.

Class A



Six students scored 60.
One scored 40 and one scored 80.

Class B



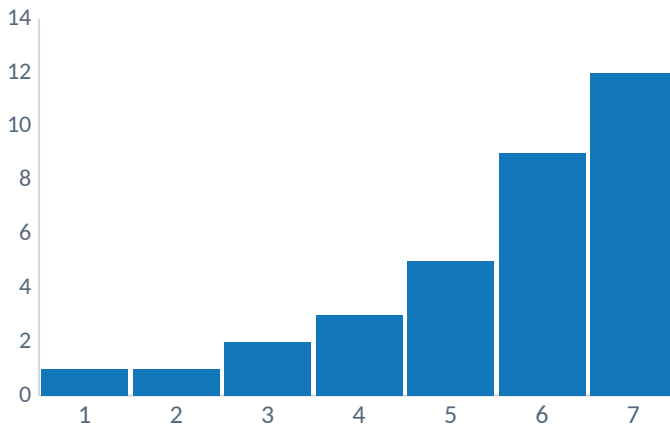
Four students scored 50.
Four scored 70. Nobody scored 60.

Class B suggests two support levels. Class A suggests a shared lesson with individual follow-up.

Skew names the direction of the long tail

The tail contains a small number of unusually low or high values.

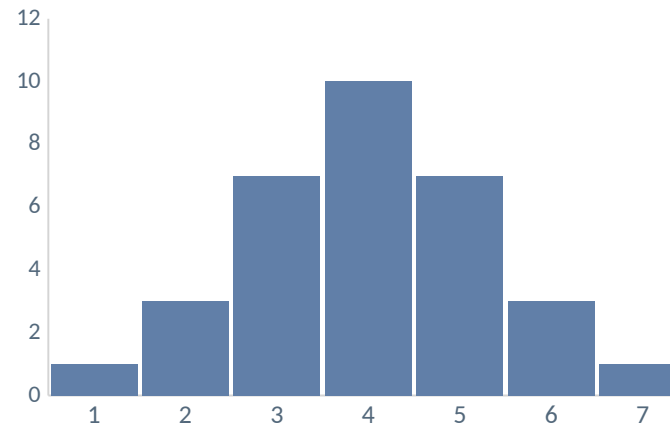
Left-skewed



Tail extends to low values

Mean tends to be below the median

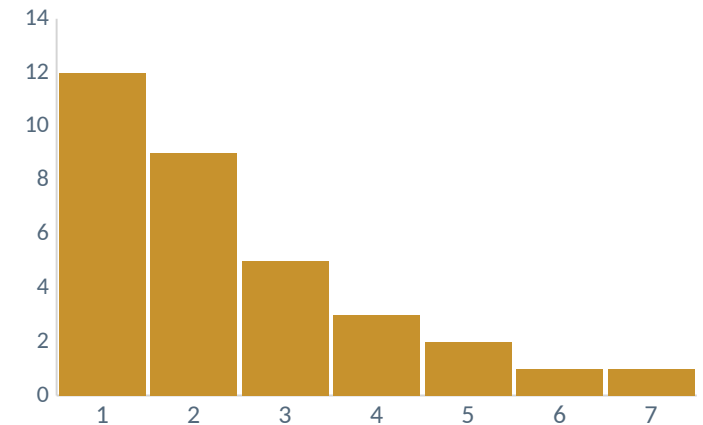
Roughly symmetric



Similar shape on both sides

Mean and median are often close

Right-skewed

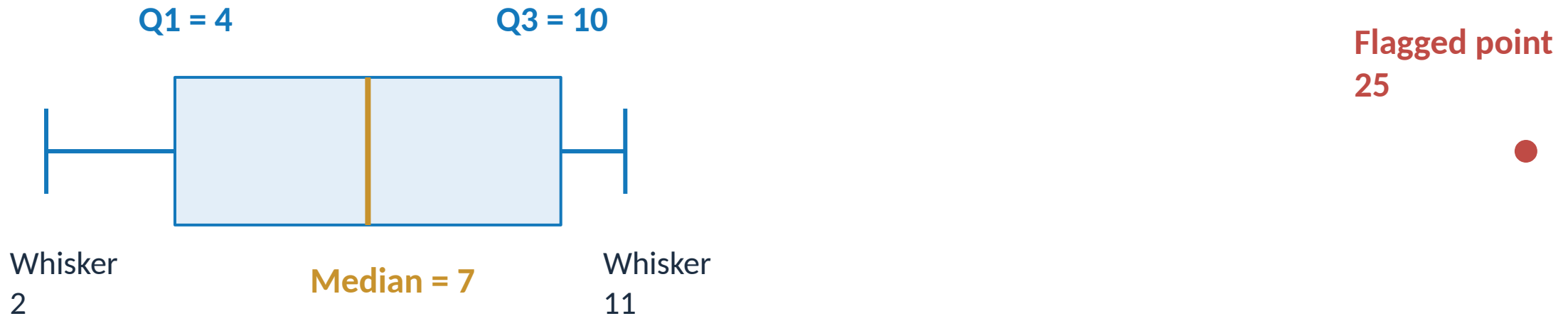


Tail extends to high values

Mean tends to be above the median

What each part of a boxplot means

Example: 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 25. Quartiles use the median of halves.



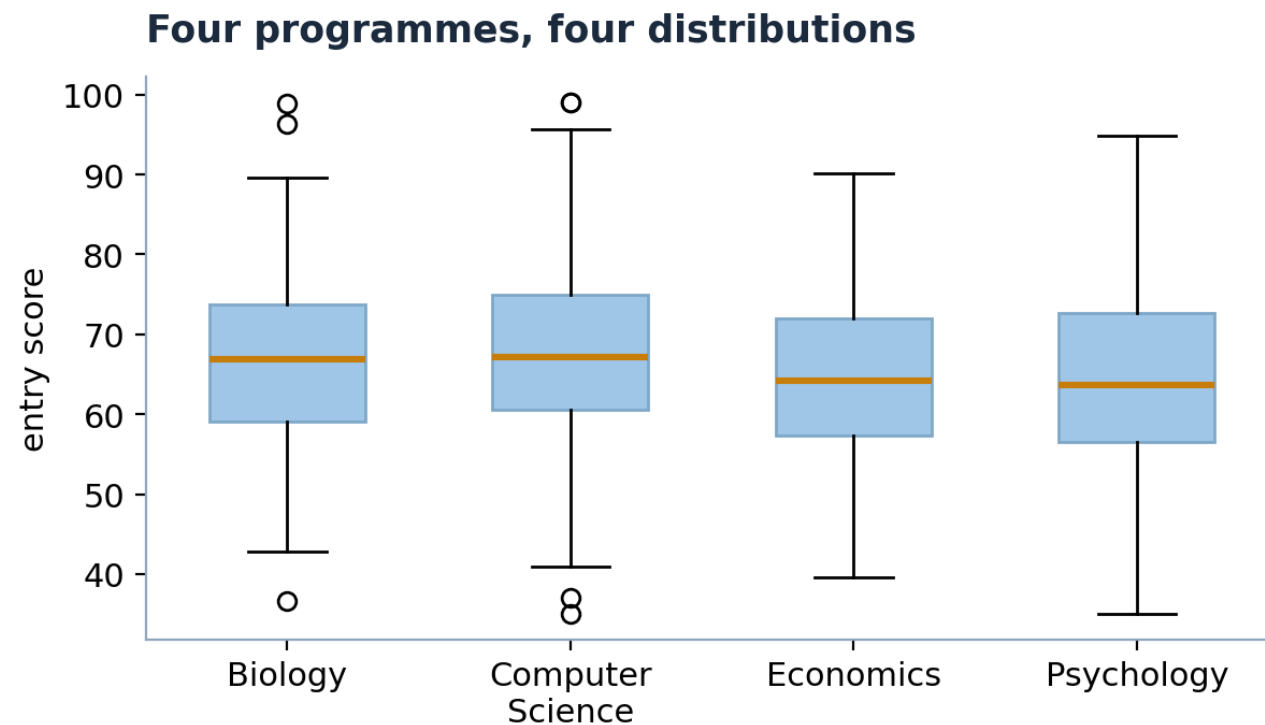
Box: Q1 to Q3, containing the middle 50%. The line inside is the median.

$IQR = 10 - 4 = 6$. Fences = $4 - 9 = -5$ and $10 + 9 = 19$.

Whiskers end at observed values inside the fences: 2 and 11. The upper fence is 19, not 11.

Entry-score patterns across four programmes

Read the median first, then the box width or height, then the whiskers and isolated points.



Biology

Median near 67. Several isolated values fall beyond the whiskers.

Computer Science

A similar median. Its box is relatively narrow, with low and high flagged values.

Economics

A lower median, near 64.
No isolated points appear in this plot.

Psychology

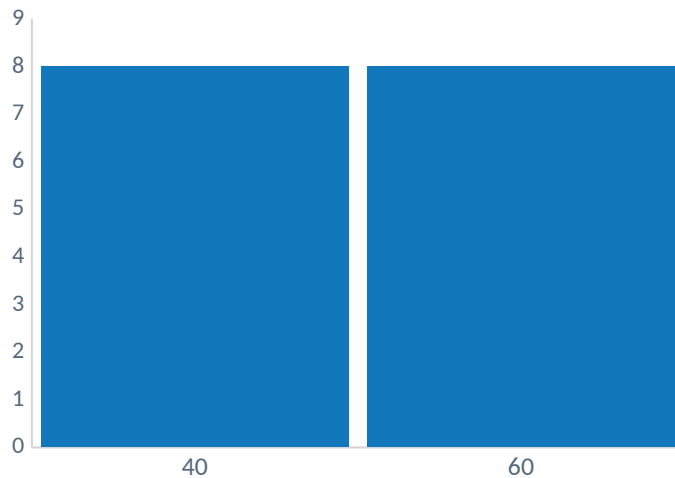
Median near 63. Its middle half spans a wider interval than CS.

All values above are approximate visual readings. A dot is an outlier candidate, not proof of an error.

Bin width can hide or reveal two groups

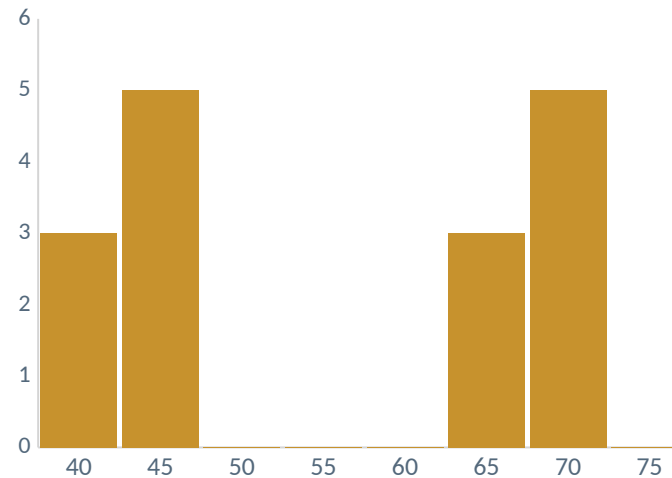
The same 16 scores appear in every plot. Only the bin width changes.

2 bins



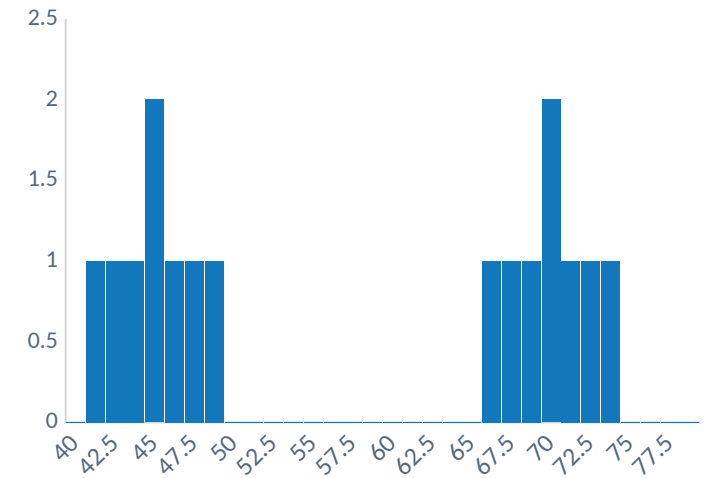
The broad bars hide the gap.

8 bins



The gap between groups is clear.

32 bins

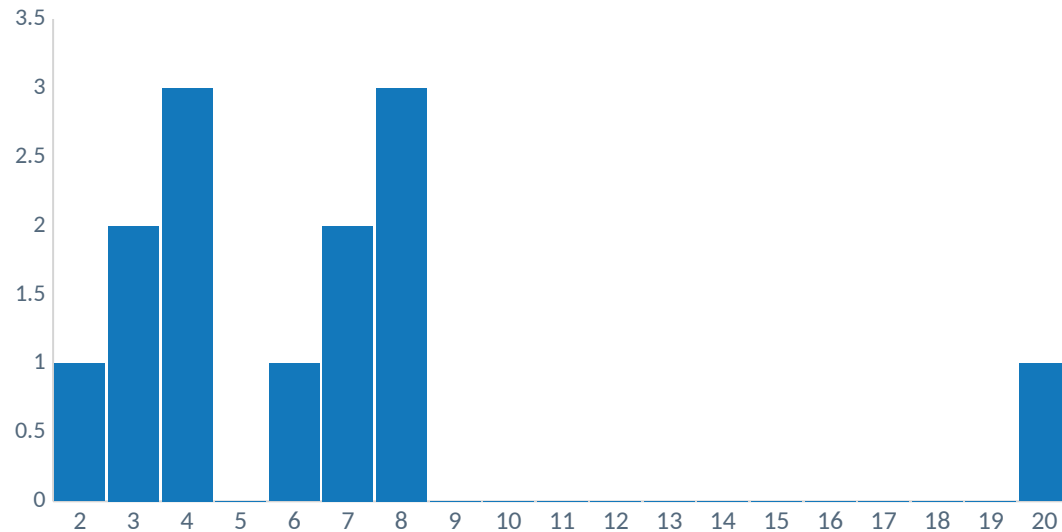


Many bars contain one value.

Scores run from 42–49 and 67–74. A useful bin choice should preserve this visible separation.

Five features of one distribution

Example scores: 2, 3, 3, 4, 4, 4, 6, 7, 7, 8, 8, 8, 20. Here $n = 13$.



Peaks at 4 and 8. A long gap separates the value 20 from the rest.

Centre

Mean = $84 \div 13 = 6.46$. Median = 6.

Spread

$s = \sqrt{[\sum(x - 6.46)^2 \div 12]} \approx 4.59$. IQR = $8 - 3.5 = 4.5$.

Skew

$g_1 = [\sum(x - \bar{x})^3/n] \div [\sum(x - \bar{x})^2/n]^{(3/2)} \approx +2.04$.

Modes and gaps

Mode = value with the highest count: 4 and 8 (3 each). No scores occur from 9 to 19.

Unusual values

Upper fence = $8 + 1.5 \times 4.5 = 14.75$. Since $20 > 14.75$, flag 20.

Two peaks ask a group question

When a distribution has distinct peaks, a single centre may lie where few observations exist.

Cohort mean = 58.8
DS100 mean = 63.1 MA110 mean = 45.4

What changed?

The courses have different grading distributions. The whole-cohort mean merges them into one number.

Better report

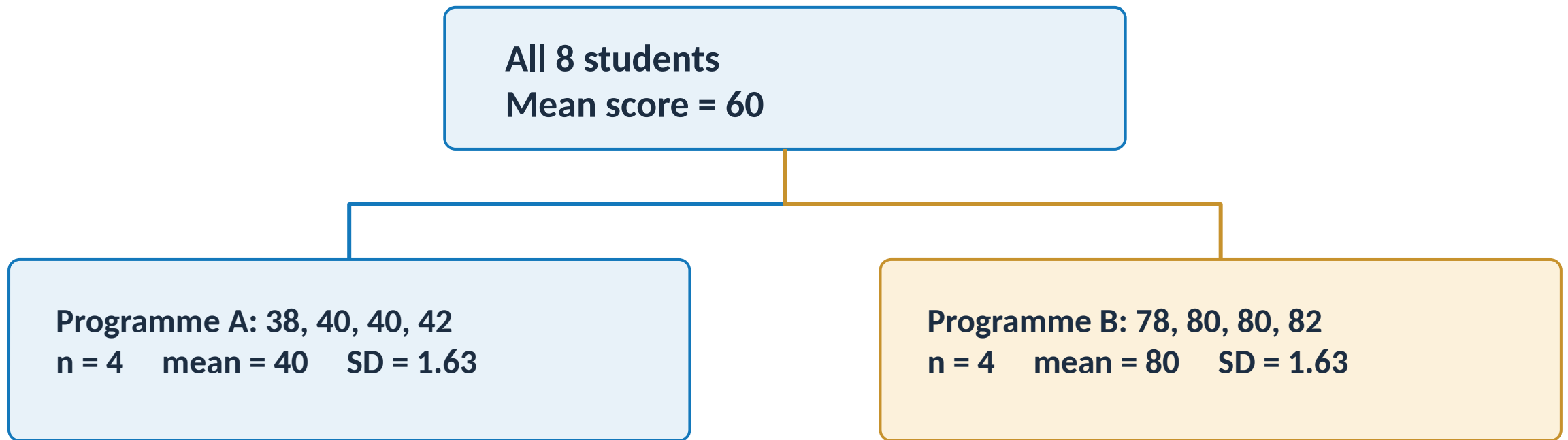
Name the grouping variable, report each group's n, centre, and spread, then explain what remains comparable.

DO NOT INFER

Different course means do not prove one course caused the difference. This is description, not causal evidence.

A group diagram reveals what the overall mean hides

Illustrative intake scores. Split the cohort by programme before interpreting the average.



The overall mean is 60, but nobody scored near 60. Each programme is tightly grouped.

Report each group's size, centre and spread. These descriptive differences do not establish a cause.

Normality is a model, not a setting

The normal distribution can be useful, but data do not become normal because software assumes it.

What a plot can do

Show an approximate shape, gaps, skew, and unusual values.

What a plot cannot do

Prove a distributional model from appearance alone.

What to do

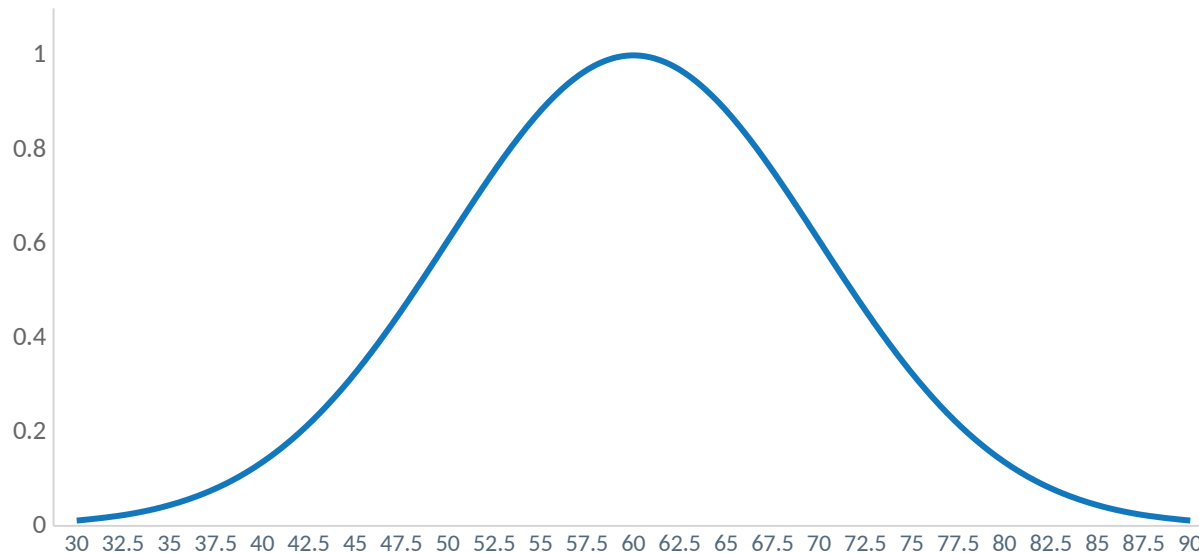
State the assumption, inspect the data, and use a method suited to the question.

PREVIEW

Later we will study when normal models help. For now, check the shape before applying a normal rule.

The 68–95–99.7 rule for a normal distribution

The rule estimates how much data lies within 1, 2 or 3 standard deviations of the mean.



Normal model

Mean $\mu = 60$, SD $\sigma = 10$

Distance from mean	Score interval	Share
$\mu \pm 1\sigma$	50 to 70	$\approx 68\%$
$\mu \pm 2\sigma$	40 to 80	$\approx 95\%$
$\mu \pm 3\sigma$	30 to 90	$\approx 99.7\%$

**For 1,000 scores following this model,
about 680 would lie from 50 to 70.**

Use this approximation for an approximately normal, bell-shaped distribution. Long tails or two peaks need a different description.

Outlier and impossible value are different claims

Both require investigation, but they call for different actions.

Unusual

A student with 6,717 platform minutes

May be valid heavy use, a duplicated log, or a different usage pattern.

Impossible

Age 305 in a student registry

Violates the data's meaning. Verify source, then correct or exclude with a documented rule.

In both cases

Keep the raw record and the decision trail

A clean dataset without a rationale is not reproducible.

NEVER

Never use a generic z-score or IQR threshold as an automatic deletion button.

Diagnose before you summarise

Choose the summary that answers the decision, then state the limitation.

Which pair would you report first for a right-skewed income distribution? Which would you report first for a nearly symmetric machine measurement?

Use a centre and a spread. Explain why the distribution makes that choice sensible.

Right-skewed income

Median and IQR.
Explain the long high-value tail.

Symmetric measurement

Mean and SD.
Check that one group and a meaningful mean exist.

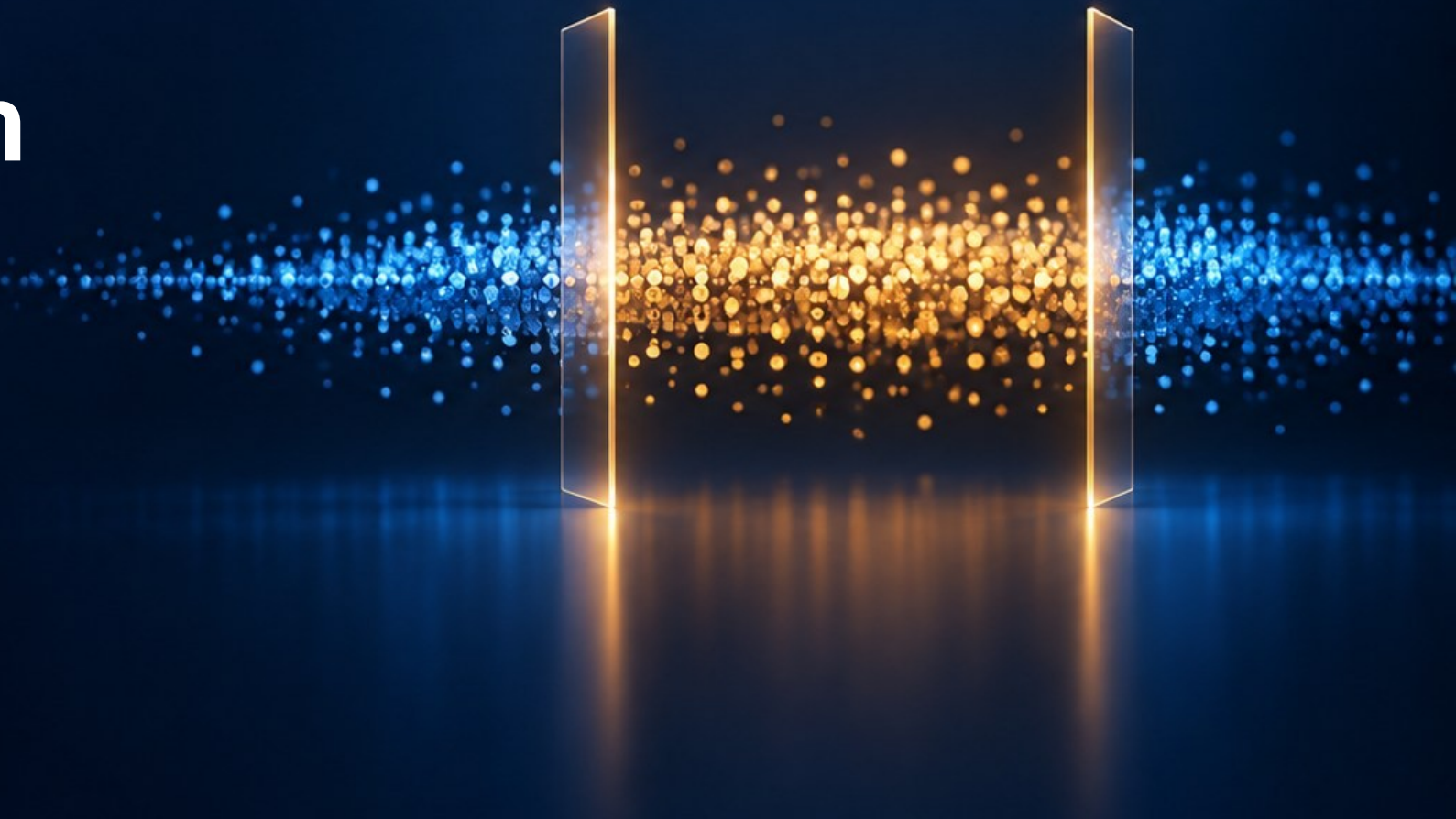
ANSWER QUALITY

A correct statistic without a reason is incomplete. The reason is the part that transfers to new data.

Standardisation

Position in units of spread

A z-score tells us how far a value lies above or below the mean of a chosen reference group.



Z-score: distance and direction from the mean

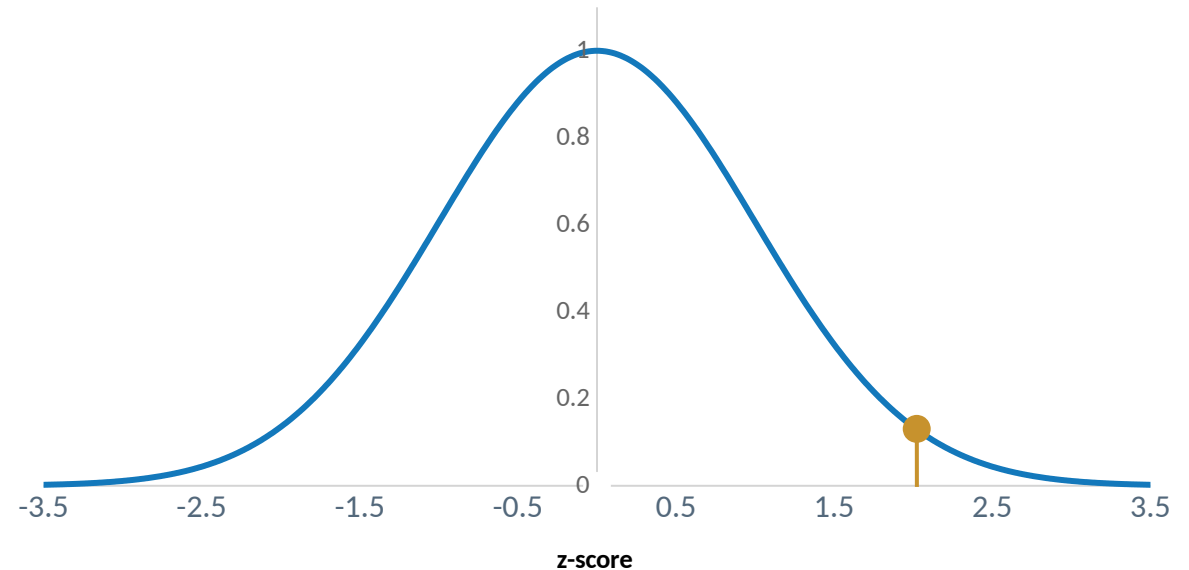
Subtract the group mean, then divide by the group standard deviation.

$$Z = \frac{x - \bar{X}}{s}$$

x: the observed value

$\bar{X}$: the reference-group mean

s: its standard deviation, with $s > 0$



Amber marker: score 88, $z = +2.03$

$$z = (88 - 65.9) \div 10.9 \approx +2.03$$

88 is 2.03 SD above the cohort mean.

The normal curve illustrates distance. The z-score calculation itself does not require normality.

What the sign and size of z tell us

The sign gives the direction. The absolute value gives the number of standard deviations.

z-score	Meaning	Example: mean 60, SD 10
0	Exactly at the group mean	Score 60
+1	One SD above the mean	Score 70
-1	One SD below the mean	Score 50
+3	Three SDs above the mean: unusually high*	Score 90
-3	Three SDs below the mean: unusually low*	Score 30

*Rarity depends on the shape. Under a normal model, about 0.3% of values lie beyond ± 3 SD.

Z-scores have no units and can exceed ± 3 . Positive does not automatically mean “good.”

Accept or reject an item using an agreed tolerance

Illustrative bottle-filling process: reference mean = 100 mL, reference SD = 2 mL.

Manufacturer's rule: accept bottles from 94 to 106 mL, inclusive.

In z units: $-3 \leq z \leq +3$. These limits are the stated specification.



Bottle A: $z = (104 - 100) \div 2 = +2$

Accept: 104 mL is inside 94–106 mL.

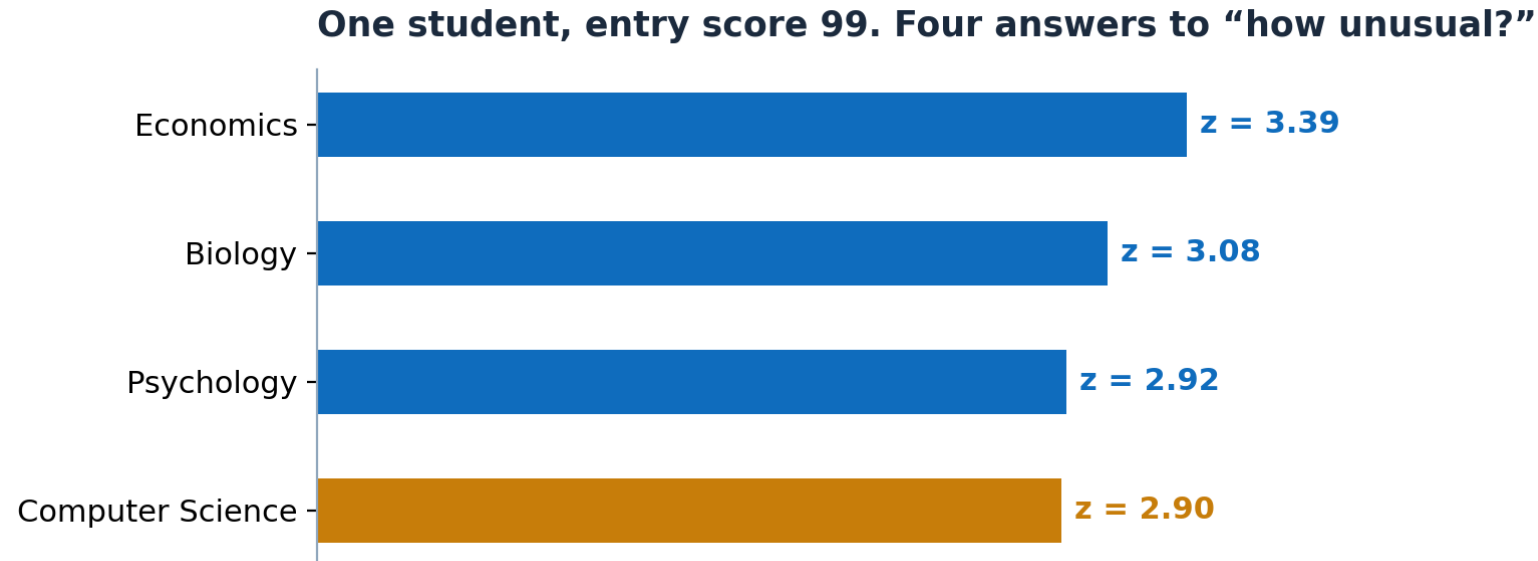
Bottle B: $z = (108 - 100) \div 2 = +4$

Reject the bottle: 108 mL exceeds 106 mL.

The specification sets acceptance, not the z-score alone. Keep both measurements in the data.

One student, entry score 99. Four answers to “how unusual?”

Student 100538 has the highest entry score in the dataset. How unusual that is depends entirely on who you compare them with.



Against Economics students, $z = 3.39$. Against their own programme, Computer Science, $z = 2.90$.

★ THE QUESTION TO ASK

A z-score is meaningless until you say: standardised against whom?

Calculate the z-score from its reference group

A z-score is a distance from one chosen mean in units of that group's standard deviation.

$$z = (99 - 65.86) \div 10.94 = +3.03$$

Plain language

Student 100538's entry score is 3.03 standard deviations above the mean of all 1,200 students.

Say the reference

The sentence is incomplete without “standardised against the whole cohort.”

DO NOT INFER

A large z-score describes position in a reference group. It does not explain why the score occurred.

The same score, interpreted against three groups

Student 100538 scored 99. Each comparison uses its own group mean and standard deviation.

Reference group	z-score	Interpretation of score 99
Whole cohort (1,200 students)	+3.03	3.03 SD above the whole-cohort mean
Computer Science	+2.90	2.90 SD above the Computer Science mean
Economics	+3.39	3.39 SD above the Economics mean

Largest relative distance: Economics

Score 99 is furthest above the Economics mean when measured in that group's SD units.

Choose the group that answers the question. A z-score does not by itself give a percentile or an admission decision.

Distance from the mean differs from percentile rank

Illustrative practice times for 10 students, sorted from lowest to highest. Focus on 9 minutes.

Sorted minutes

1	2	3	4	5	6	7	8	9	50
---	---	---	---	---	---	---	---	---	----

9 of the 10 values are at or below 9 minutes

Z-score: distance

Mean = 9.5 min, sample SD $\approx$ 14.46 min

$$z = (9 - 9.5) \div 14.46 \approx -0.03$$

9 minutes is slightly below the mean.

Empirical percentile: rank

$$\text{Percentile rank} = 9 \div 10 \times 100 = 90\%$$

9 minutes is at the 90th percentile using the “at or below” definition.

The value 50 pulls the mean upward. A time below the mean can still exceed most observations.

Standardise measurements, not arbitrary codes

A formula can produce a number even when the number has no meaning.

Entry score

0–100 points: z-score can be meaningful

The distance in points has a defined interpretation.

Platform minutes

A meaningful unit, but inspect shape first

A z-score may not give the clearest typical-case description for a long-tailed variable.

Programme code

1–4: do not standardise

The arithmetic exists but the numerical distance between categories does not.

WEEK 1 RETURNS

Measurement level decides whether an operation is meaningful, not whether a library function accepts it.

Four questions, in order

01

What decision does this number feed?

Budget totals need the mean. “Typical” needs the median.

02

What shape is the column?

Plot the data first. The shape may change which summary makes sense.

03

Could this be two groups?

If the histogram has two peaks, report the groups separately when appropriate.

04

What am I discarding, and does it matter?

Name it in a sentence. That sentence is what separates a report from a number.

Course rule two, restated: a number without a sentence interpreting it is not an answer. Question 04 is that sentence.

FROM WEEK 3 ONWARD

Every summary you submit carries a sentence naming what it discarded. It is marked.

Everything from today, in eight lines

```
students["entry_score"].mean()          # 65.86
students["entry_score"].median()        # 65.90
students["entry_score"].std()           # 10.94
students["entry_score"].quantile(.25)   # 58.3
students["entry_score"].skew()          # -0.01

students["entry_score"].describe()
students["entry_score"].hist(bins=30)

students.groupby("programme")["entry_score"].agg(
    ["count", "mean", "median", "std"])
```

Two warnings

`describe()` will happily summarise `student_id`. It is still meaningless.

numpy's `std()` divides by `n`; pandas divides by `n - 1`. On the same column they disagree. Use pandas, or pass `ddof=1`.

★ DO IT NOW

Run the `groupby` line. Which programme has the widest spread?

Profile one numeric column before modelling

A short, reproducible profile catches surprises before they become model features.

```
x = students['entry_score']  
  
x.agg(['count', 'mean', 'median', 'std', 'min', 'max'])  
x.quantile([.25, .50, .75])  
x.hist(bins=20)
```

Read the output

Count and quantiles answer different questions from mean and SD. Keep all of them in view.

Then interpret

Write one sentence about centre, one about spread, and one about shape or limitation.

REPRODUCIBILITY

The code is short. The explanation is the assessed work.

A group-wise profile finds hidden structure

Aggregate after deciding what one row should mean, then report the group definition.

```
(enrolments.groupby('course_code').final_grade
 .agg(n='count', mean='mean', median='median', sd='std')
 .sort_values('mean'))
```

What this changes

The overall final-grade mean of 58.8 becomes a set of course-level distributions that can be inspected honestly.

What it cannot do

It cannot establish that a course caused the difference. It only makes the structure visible.

RULE

Do not choose subgroups only after seeing an interesting result. State why the group belongs in the analysis.

A defensible reporting sentence

The reader should not need to reconstruct the context from your code.

Among 1,200 students, entry scores had a mean of 65.9 points and a standard deviation of 10.9 points. The distribution was nearly symmetric; the middle half of scores lay from 58.3 to 73.6 points.

It states population, n, unit, centre, spread, shape, and an additional robust summary.

What it avoids

“Average score = 65.9.”
No unit, n, spread, or shape.

What remains

It still does not prove a cause, explain the scores, or describe missing data in another column.

WRITING RULE

A number without a sentence interpreting it is not an answer.

Five mistakes, all of them common

Reporting a mean with no spread

The single most common fault in student reports, and the easiest to fix.

Using the 68/95 rule on skewed data

It assumes a bell shape. Check the shape first.

Reporting the mean of an ordinal scale

“Mean satisfaction 3.4” treats unequal gaps as equal. Report the distribution too.

Summarising before plotting

Two lumps and one lump give the same mean. You cannot tell without looking.

Standardising without naming the reference group

$z = 2.9$ or $z = 3.39$ for the same student, depending on who you compared them with.

★ ONE HABIT PREVENTS FOUR OF THE FIVE

Plot the column before you summarise it.

What you can do now

A summary is compression

and you are responsible for what you discarded

Mean and median answer different questions

when they disagree, that disagreement is information

A centre needs a measure of spread

The SD of 10.9 explains how much entry scores differ around 65.9.

Two numbers assume one lump

plot it, or you are guessing

A z-score needs a reference group

the same student is $z = 2.90$ or $z = 3.39$ depending on the comparison

Last week: what a number is allowed to mean. This week: what it leaves out. Next week: whether you can trust it at all.

THE HABIT

Never report a centre without a spread — and never report either without looking at the shape.

Choose a summary for platform use

The Dean asks: “What does a typical student’s platform use look like?”

Total platform minutes per student have mean 1,592, median 1,426, a long right tail, and 23 values above the 1.5 IQR upper fence.

Write a two-sentence answer for the Dean. Which centre and spread describe the typical student? What would you report separately for workload planning?

Typical student

Median 1,426 and IQR.
Explain the right-skewed distribution.

Capacity planning

Mean or upper quantiles may matter.
Explain why high-use students affect resource demand.

TRANSFER

The statistic follows the decision. There is no universally “best” centre.

Four short answers before you leave

- 1 The mean of minutes on platform is 1,592 and the median is 1,426. Which would you report as 'typical', and why?
- 2 `entry_score` has mean 65.9 and sd 10.9. Between which values do about two-thirds of students sit?
- 3 Two courses both have a mean grade of 58. Name one thing that could still make them completely different.
- 4 A student has $z = +2.1$. State what that means, and name the one piece of information you need before you can interpret it.

★ HAND IN

On your way out. Bring your answers to Lab 2.

Summaries by hand, then in code

By hand first

Eight numbers, on paper: mean, median, sd. Then check against pandas.

Deliverable

One notebook, plus a 150-word note on one column whose mean you would not report alone.

In the lab you will

- Compute mean, median and standard deviation by hand for a small set, then verify in pandas
- Plot four columns and describe the shape of each in one sentence
- Find one column where the mean and median disagree, and explain why
- Standardise one column and state explicitly which group it was standardised against

Carry this rule: never report a centre without a spread, and plot before you summarise.